RIG-SMOOTH MORPHISMS ARE CLOSED UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the counted statement is the author lemma environment labelled lemma-composition-rig-smooth; the proof environment immediately following it is its own proof

Editorial note. The label lemma-composition-rig-smooth names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the WAdm$\widehat{}$Noeth-level composition lemma for rig-smooth arrows (lemma-composition-rig-smooth-continuous of the same chapter)
- the Formal Spaces Noetherian variant remark for composition (formal-spaces-remark-composition-variant-Noetherian)

Source reference. Original remark 0GBK: sources/stacks-chow/tags.txt, source lines 20375–20375. Prerequisite pointer invoked at 4716; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GCM: sources/stacks-restricted/pinned-original-source.tex, source lines 4394–4394. Prerequisite pointer invoked at 4717; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of locally Noetherian formal algebraic spaces over S. If f and g are rig-smooth, then so is $g \circ f$.*

Proof. By Formal Spaces, Remark 0GBK this follows from Lemma 0GCM. □

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